P2P-CD-01

Peer-to-Peer Capability Discovery with Trust Gates

Specification Draft — Version 0.4

Status: Work in Progress | Category: Proposed Standard

Abstract

This document specifies a protocol for capability discovery between peers over connected sessions, with access control gates built into the protocol core. The protocol defines a CBOR-based wire format for capability advertisement through the discovery manifest, a hash-driven session state machine, role-aware capability negotiation with mutual active-set confirmation, and a peer cache mechanism for efficient re-connection handling. Trust credential schemes and transport bindings are defined in appendices.

1. Introduction

Capability discovery between peers is underspecified in existing P2P protocols. Protocols such as libp2p identify, XMPP service discovery, and WebRTC SDP each solve parts of the problem in incompatible ways, with access control treated as an application-layer concern. This produces interoperability failures and forces implementations to reinvent authorization logic.

This specification defines a minimal, interoperable protocol for capability discovery that treats access control as a first-class protocol primitive. The core design goals are:

Trust-gated: access control is enforced before capability activation, not after.

Asymmetry-aware: peers independently declare provide and consume postures; role matching is explicit.

Change-reactive: any change to a peer's capability state triggers re-exchange via a hash-driven state machine.

Minimal: the normative core defines the smallest surface required for interoperability; policy, credentials, and transport bindings are appendices.

1.1 Out of Scope

The following are explicitly out of scope for this specification:

Peer discovery transport (how peers become aware of each other)

Identity provider integration and credential issuance

Behavioral reputation and scoring

Cryptographic algorithm mandates (algorithm agility is required; specific algorithms are not mandated)

Application-level capability semantics beyond namespace grammar

2. Terminology

The key words MUST, MUST NOT, SHOULD, SHOULD NOT, and MAY in this document are to be interpreted as described in RFC 2119.

Peer: A node that implements this protocol and participates in sessions.

Session: A mutually authenticated connection between exactly two peers over which capability exchange occurs.

Capability: A named, versioned, role-tagged function that a peer can provide to or consume from other peers. Capabilities are the atomic unit of negotiation. Each capability is declared using the namespace grammar defined in §4.4.

Discovery Manifest: The complete set of capability declarations, control parameters, and metadata advertised by a peer in a single exchange message. The canonical wire encoding is CBOR (§5).

Personal Hash: A hash computed by a peer over the deterministic CBOR serialization of its own discovery manifest (§4.5, §5.2). Each peer owns and maintains exactly one personal hash.

Classification: A group reference assigned to a peer that governs which capabilities and scope parameters are applied to that peer's session. Classification is resolved per-capability: a single peer MAY receive different classifications for different capabilities within the same session.

Scope: The set of constraints applied to a capability for a given peer. Scope parameters are registry-extensible; the core protocol defines rate_limit and ttl.

Trust Gate: A gating predicate evaluated per-capability against a peer's resolved classification before that capability is activated. A trust gate produces ALLOW or DENY.

PEER_VISIBLE: A pre-session disposition. The peer is known to exist. No session is implied and no protocol obligations arise.

ACTIVE: A session state in which at least one capability was mutually negotiated, confirmed, and activated.

NONE: A terminal session outcome in which no capabilities were negotiated. Always results in session close.

DENIED: A pre-session disposition in which a previously seen peer hash is known to produce NONE. Negotiation is skipped. No transport connection is opened.

Active Set: The set of capabilities that both peers have confirmed as activated for a given session. The active set is the intersection of both peers' CONFIRM messages (§7.2).

3. Peer Discovery

Peer discovery is the process by which a peer becomes PEER_VISIBLE to another. This specification does not define the discovery transport. Discovery MAY occur over any broadcast or unicast medium including but not limited to local wireless broadcast, multicast DNS, DHT lookup, or rendezvous server.

3.1 Discovery Broadcast

A discovery broadcast MUST include the peer_id and personal_hash fields. A discovery broadcast MAY include the full discovery manifest. If the manifest is

omitted, the receiving peer knows the peer exists and has a hash fingerprint, but MUST initiate a session to obtain the full manifest.

Receipt of a valid discovery broadcast MUST cause the receiving peer to record the sender as PEER_VISIBLE and store the sender's peer_id, personal_hash, and last_seen timestamp in the peer cache (§9).

When the full manifest is included in the discovery broadcast, it enables a receiving peer to evaluate compatibility and apply auto-DENY before any handshake occurs, reducing unnecessary session overhead. It also enables application-level use cases where visibility of a peer's capability intent prior to negotiation is a feature.

When the full manifest is omitted (lightweight discovery), auto-deny based on hash comparison still functions (the hash is always present), but pre-session compatibility filtering is not possible. Implementations that require capability confidentiality prior to authentication SHOULD use lightweight discovery.

3.2 Discovery Is Not a Session

A peer becoming PEER_VISIBLE does not initiate capability negotiation. A peer MAY remain PEER_VISIBLE indefinitely without a session being established. No protocol obligations arise from PEER_VISIBLE state alone.

4. Discovery Manifest Format

The discovery manifest is the atomic unit of capability advertisement. It is transmitted during session establishment (OFFER messages) and, optionally, during discovery broadcasts. When transmitted, the manifest MUST be sent in full; partial or delta manifests are not permitted in this version of the protocol.

4.1 Top-Level Fields

The discovery manifest contains the following top-level fields:

```
discovery_manifest {
    protocol_version : <unsigned integer>
    peer_id          : <cryptographic identity handle (byte string)>
```

```
    sequence_num     : <monotonically increasing unsigned integer>
    capabilities     : [ capability_declaration, ... ]
    personal_hash    : <hash of manifest state (byte string)>
    hash_algorithm   : <algorithm identifier (text string)>
}
```

All fields listed above are MUST-implement. A manifest missing any field MUST be rejected.

The protocol_version field identifies the version of this specification. For this document, the value MUST be 1. A peer that receives a manifest with a protocol_version it does not support MUST reject the manifest and SHOULD send a CLOSE message indicating unsupported version. Future revisions of this specification will increment this value. Peers SHOULD NOT attempt to negotiate downward; version negotiation is not defined in this version of the protocol.

The sequence_num field MUST increment with every new manifest issued by a peer. A receiver that observes a sequence_num equal to or less than the last seen sequence_num from that peer MUST treat the manifest as a replay and discard it.

4.2 Capability Declaration

Each entry in the capabilities array is a capability declaration:

```
capability_declaration {
    name               : <namespace string (text string)>
    role               : PROVIDE | CONSUME | BOTH
    mutual             : <boolean, default false>
    classification     : <group_ref (text string) | credential_blob (byte string)
    scope              : scope_params
    applicable_scope_keys : <array of unsigned integers, optional>
}
```

```
scope_params {
    rate_limit       : <unsigned integer, requests per second, 0 = unlimited>
    ttl              : <unsigned integer, seconds, 0 = no expiry>
}
```

The role field MUST be present on every capability declaration.

The mutual field MUST be set to true for capabilities that are intended to activate under a BOTH + BOTH role match (§7.4). If omitted, the default value is false.

The classification field MAY be omitted. If omitted, the receiving peer MUST apply its local default classification policy. Classification is per-capability: a peer MAY present different credentials or group references for different capabilities within the same manifest. When the classification carries a cryptographic credential, it SHOULD be encoded as a COSE structure (§5.4) for consistency with the wire format.

The applicable_scope_keys field is an optional array of scope parameter integer keys that are meaningful for this capability. If present, a receiver MUST enforce only the listed scope parameters and MUST ignore all others, even if values are present in scope-params. If absent, the receiver MUST fall back to the capability's specification document.

The scope_params fields are registry-extensible. The core protocol defines rate_limit and ttl. Additional scope parameters MAY be registered in the P2P-CAPDISC Scope Parameter Registry (§11). Capability definitions registered in the P2P-CAPDISC Well-Known Capabilities Registry MUST declare which scope parameters are applicable; scope parameters that are not applicable to a given capability MUST be ignored by the receiving peer.

4.3 Role Semantics

The role field declares a behavioral contract for the capability. The following roles are defined:

PROVIDE: The peer commits to accept incoming requests for this capability from matched peers. A PROVIDE peer acts as the server for this capability. If a PROVIDE peer cannot fulfill a capability after activation (e.g., due to resource exhaustion), it MUST rebroadcast with the capability removed or its scope modified (§8).

CONSUME: The peer declares intent to initiate requests against this capability on matched peers. A CONSUME peer acts as the client for this capability.

BOTH: The peer commits to both accept incoming requests and initiate outgoing requests simultaneously. A BOTH peer acts as both server and client. Capabilities declared with role BOTH that are intended to activate under a BOTH + BOTH match MUST also set mutual: true.

The role field is a binding declaration, not merely a statement of intent. A peer that advertises a role for a capability MUST be prepared to fulfill that role upon activation.

4.4 Namespace Grammar

Capability names MUST conform to the following grammar:

```
name         :=  org '.' scope '.' capability '.' version
org          :=  label ( '.' label )*
label        :=  [a-z] [a-z0-9-]* [a-z0-9]
scope        :=  [a-z] [a-z0-9]*
capability   :=  [a-z] [a-z0-9]*
version      :=  [0-9]+
```

The org segment is a reverse-domain-style organizational prefix that prevents namespace collisions across independent implementations.

The scope segment SHOULD identify the functional layer or domain that the capability addresses. For capabilities in the reserved core.* namespace, the following scope segments are defined:

session — capabilities concerning the health and verification of the session itself (liveness, timing, attestation).

network — capabilities concerning network topology, addressing, and reachability (endpoint reflection, relay, peer exchange).

data — capabilities concerning data movement between peers (streams, blobs, RPC, events).

bridge — (reserved for future use) capabilities concerning interoperation with non-P2P protocols.

Capability definitions registered under core.* MUST use one of the defined scope segments. Application-defined capabilities (non-core.*) MAY use any scope segment conforming to the grammar.

The namespace prefix "core" is RESERVED for capabilities defined by this specification and its normative extensions. Implementations MUST NOT register capabilities under "core.*" without registration in the P2P-CAPDISC Well-Known Capabilities Registry (§11).

Examples of conformant capability names: core.session.heartbeat.1, core.data.stream.1, core.network.relay.1, com.example.media.transcode.2, org.freedesktop.desktop.share.1.

Version is a required component of the name and represents a breaking change to the capability's semantics or wire format. Two capability names that differ only in version (e.g., core.data.blob.1 and core.data.blob.2) MUST be treated as distinct capabilities for the purposes of intersection computation. Non-breaking changes SHOULD be handled through scope parameters or capability-specific extension fields, not version increments.

4.5 Personal Hash Computation

The personal hash is computed over the deterministic CBOR serialization (§5.2) of the peer's manifest state:

```
personal_hash = hash_algorithm;H(canonical_cbor(peer_id, sequence_num, sorted(cap
```

Capabilities MUST be sorted lexicographically by name before serialization and hashing. Implementations that do not sort capabilities will produce non-deterministic hashes and MUST be considered non-conformant.

The sequence_num MUST be included in the hash computation to ensure that replayed manifests with identical capability sets but stale sequence numbers produce different hashes.

The hash input MUST be the deterministic CBOR encoding (§5.2) of the fields being hashed. This ensures that two conforming implementations with the same logical state always produce the same hash value.

The hash_algorithm field MUST be prepended in the manifest.

Note: This specification does not mandate a specific hash algorithm. Implementations SHOULD select from algorithms registered in the IANA Named Information Hash Algorithm Registry. The algorithm identifier in the manifest MUST use the IANA registered name. Implementations SHOULD support SHA-256 as a baseline for interoperability.

4.6 Capability Definition Requirements

A capability definition registered in the P2P-CAPDISC Well-Known Capabilities Registry MUST include all of the following:

1. **Name** — fully qualified namespace name conforming to §4.4.

2. **Role** — which role values are valid for this capability (PROVIDE, CONSUME, BOTH, or a subset).

3. **Mutual** — whether BOTH + BOTH activation requires mutual: true; if omitted, default is false.

4. **Capability-specific parameters** — CDDL definitions for any parameters beyond rate_limit and ttl, carried in the capability's scope-params extension keys.

5. **Scope applicability** — which core scope parameters (rate_limit, ttl) are applicable, and which MUST be ignored.

6. **Message types** — integer values for all messages the capability defines, with CDDL schemas conforming to the capability message envelope (§5.3.7).

7. **Parameter negotiation rules** — for each capability-specific parameter, the reconciliation rule (most-restrictive-wins, provider-takes-precedence, or custom). Non-numeric parameters MUST define explicit reconciliation.

8. **Activation exchange** — if the capability requires a message exchange upon entering the active set, the sequence MUST be defined here, including timeout behavior.

9. **Conformance level** — whether the capability is Mandatory, Recommended, or Optional for each conformance level (§D).

A capability definition SHOULD also include:

10. **Transport requirements** — if the capability requires specific transport properties (e.g., ordered delivery, datagram support).

11. **Security considerations** — capability-specific security implications.

5. Wire Format

This section defines the canonical wire encoding for all protocol messages. A single mandatory encoding ensures that two independent implementations produce

identical byte sequences for the same logical message, which is essential for the hash-driven state machine defined in §4.5.

5.1 Canonical Encoding: CBOR

All protocol messages — discovery broadcasts, OFFER messages, CONFIRM messages, and CLOSE messages — MUST be encoded using CBOR (Concise Binary Object Representation) as defined in STD 94 (RFC 8949).

CBOR is mandated for the following reasons specific to this protocol:

Native binary byte strings: peer identities, hash digests, and credential blobs are binary data. CBOR encodes these directly without base64 overhead or encoding ambiguity.

Deterministic serialization: the personal hash (§4.5) requires that two implementations produce identical byte sequences from the same logical manifest. CBOR's deterministic encoding profile (§5.2) provides this guarantee.

Self-describing: CBOR is schema-free. A decoder can parse any conforming message without prior knowledge of the capability definitions it contains. This is essential for the registry-extensible scope parameters and decentralized namespace defined in this specification.

IETF alignment: CBOR is an Internet Standard (STD 94) with IANA-registered extension tags, a formal schema language (CDDL, RFC 8610), and a companion signing and encryption specification (COSE, RFC 9052). These integrate directly with the protocol's trust gate and classification mechanisms.

5.2 Deterministic Encoding Profile

All CBOR encodings produced by this protocol MUST conform to the CBOR Core Deterministic Encoding Requirements defined in RFC 8949 §4.2.1. In summary:

Integers MUST be encoded in the smallest possible number of bytes.

Map keys MUST be sorted in bytewise lexicographic order of their encoded form.

Indefinite-length items MUST NOT be used.

Floating-point values MUST use the smallest precision that preserves the value (float16 if sufficient, then float32, then float64). For this protocol, all scope

parameters are integers; floating-point values SHOULD NOT appear in protocol messages.

These requirements ensure that two conforming implementations encoding the same logical manifest will produce byte-identical output. This is the foundation of the hash-driven state machine: if two manifests are logically equal, their deterministic CBOR encodings are identical, and therefore their personal hashes are identical.

5.3 CDDL Schema Definitions

The following schemas are defined using the Concise Data Definition Language (CDDL) as specified in RFC 8610. These schemas are normative. A message that does not conform to the applicable schema MUST be rejected.

5.3.1 Discovery Manifest

```
discovery-manifest = {
    1 : uint,                    ; protocol_version (MUST be 1)
    2 : bstr,                    ; peer_id
    3 : uint,                    ; sequence_num
    4 : [+ capability-declaration],  ; capabilities
    5 : bstr,                    ; personal_hash
    6 : tstr,                    ; hash_algorithm (IANA registered name)
}
```

Map keys are integers to minimize encoded size. Implementations MUST use the integer keys as specified; text-string keys are not permitted in the canonical encoding.

5.3.2 Capability Declaration

```
capability-declaration = {
    1 : tstr,                    ; name (namespace grammar, §4.4)
    2 : role,                    ; role
    ? 3 : bool,                  ; mutual (default: false)
    ? 4 : classification,        ; classification
    ? 5 : scope-params,          ; scope
    ? 6 : [* uint],              ; applicable_scope_keys
}

role = &(
```

```
    provide: 1,
    consume: 2,
    both:    3,
)

classification = tstr / bstr     ; group_ref (tstr) or credential_blob (bstr)

scope-params = {
    ? 1 : uint,                  ; rate_limit (requests/sec, 0 = unlimited)
    ? 2 : uint,                  ; ttl (seconds, 0 = no expiry)
    * int => any,                ; registry-extensible additional params
}
```

The scope-params map is open-ended: the core protocol defines keys 1 (rate_limit) and 2 (ttl). Additional keys are allocated via the P2P-CAPDISC Scope Parameter Registry (§11). A receiver that encounters an unrecognized scope parameter key MUST ignore it.

5.3.3 OFFER Message

```
offer-message = {
    1 : uint,                    ; message_type (MUST be 1 for OFFER)
    2 : discovery-manifest,      ; the sender's full manifest
}
```

The change_flags field is advisory and non-normative. Receiving peers MUST still perform full re-evaluation regardless of change_flags contents. Integer values are defined in the change-flag enumeration above.

5.3.4 CONFIRM Message

```
confirm-message = {
    1 : uint,                    ; message_type (MUST be 2 for CONFIRM)
    2 : bstr,                    ; personal_hash (from sender's preceding OFFER)
    3 : [* tstr],                ; active_set (sorted capability names)
    ? 4 : {* tstr => scope-params},  ; accepted_params (per-capability)
}
```

The active_set array contains the fully qualified capability names that the sending peer considers activated. Entries MUST be sorted lexicographically. An empty array

indicates no capabilities were matched.

The accepted_params map is OPTIONAL. When present, it carries the scope parameters that the sending peer has accepted or counter-proposed for each activated capability, keyed by capability name. If accepted_params is omitted, the receiver MUST treat the scope parameters from the OFFER as accepted without modification. If accepted_params is present for a given capability and the values differ from the OFFER, the receiving peer MUST use the more restrictive value for each numeric parameter (the lower of the two rate_limit values; the shorter of the two ttl values). Non-numeric parameter reconciliation is capability-specific and MUST be defined in the capability's specification.

5.3.5 CLOSE Message

```
close-message = {
    1 : uint,                   ; message_type (MUST be 3 for CLOSE)
    2 : bstr,                   ; personal_hash (sender's current hash)
    ? 3 : uint,                 ; reason_code
}
```

Reason codes are registry-extensible. The following are defined by this specification:

```
reason-code = &(
    normal:              0,      ; orderly shutdown
    no-match:            1,      ; NONE outcome
    auth-failure:        2,      ; HANDSHAKE failed
    version-unsupported: 3,      ; protocol_version not supported
    timeout:             4,      ; exchange timeout
    error:               255,    ; unspecified error
)
```

5.3.6 Message Framing

The message_type field (key 1) in each message allows a receiver to dispatch on the first decoded field. The defined message types are:

```
message-type = &(
    offer:   1,
    confirm: 2,
```

```
    close:   3,
)
```

Message type values are allocated in the following ranges:

Range	Allocation	Authority
1–3	Protocol messages (OFFER, CONFIRM, CLOSE)	This specification
4–31	Reserved for `core.*` capabilities	P2P-CAPDISC Message Type Registry
32–255	Application-defined	First-come, no registration required
256+	Extended application-defined	First-come, no registration required

Capability definitions registered under core.* MUST allocate message types from the 4–31 range. Application-defined capabilities MUST NOT use message types in the 1–31 range. Two capabilities that allocate the same message type value MUST NOT both appear in the same active set; implementations MUST detect this conflict during intersection computation and exclude the later-registered capability.

All messages are self-contained CBOR maps. When transmitted over a stream transport (TCP, QUIC stream, WebSocket), each message MUST be preceded by a 4-byte unsigned big-endian length prefix indicating the number of CBOR bytes that follow. Over datagram transports (UDP broadcast), each datagram contains exactly one message with no length prefix.

5.3.7 Capability Message Envelope

All messages defined by capability specifications — whether registered under core.* or application-defined — MUST conform to the following envelope structure:

```
capability-message = {
    1 : uint,          ; message_type (from allocated range)
    * uint => any,     ; capability-defined fields (keys >= 2)
}
```

Key 1 MUST contain the message type. Capability-specific fields MUST use integer keys >= 2. A receiver that encounters a message with an unrecognized message type

MUST ignore the message and MUST NOT close the session.

The envelope ensures that any conforming decoder can extract the message type from any capability message without knowledge of the capability's schema, enabling generic message dispatch.

5.4 Credential Encoding with COSE

When the classification field in a capability declaration carries a cryptographic credential (as opposed to a plain group_ref text string), the credential SHOULD be encoded using COSE (RFC 9052). This ensures that credential verification uses the same binary encoding as the rest of the protocol, avoiding format conversion between CBOR manifests and JSON-based credential formats.

The following COSE structures are applicable:

COSE_Sign1 (CBOR tag 18): A single-signer signed credential. Suitable for JWTs converted to CWT (CBOR Web Token, RFC 8392) format and for signed classification assertions.

COSE_Mac0 (CBOR tag 17): A MAC-authenticated credential for symmetric-key environments.

COSE_Encrypt0 (CBOR tag 16): An encrypted credential for environments requiring classification confidentiality.

A receiving peer MUST verify the COSE structure before using the claims within it for classification resolution (§6.2). The verification key is determined by the receiving peer's local trust policy and is out of scope for this specification.

Implementations MAY also accept non-COSE credential formats (JWT, W3C Verifiable Credentials, Macaroons) as described in Appendix A. However, COSE is RECOMMENDED for new deployments to maintain a single serialization format throughout the protocol stack.

5.5 JSON Interoperability

For debugging, logging, and interoperability with web-based systems, implementations MAY support JSON (RFC 8259) as an alternative encoding for display and diagnostic purposes. However:

The canonical encoding for hash computation MUST be deterministic CBOR (§5.2). Implementations MUST NOT compute personal hashes over JSON-encoded data.

The canonical encoding for wire transmission MUST be CBOR. Implementations MUST NOT transmit JSON-encoded messages in protocol exchanges.

When JSON is used for diagnostic output, the CBOR diagnostic notation defined in RFC 8949 §8 is RECOMMENDED for fidelity. Implementations MAY also produce standard JSON by converting CBOR byte strings to base64url and CBOR integer map keys to their string equivalents.

6. Trust Gates and Peer Classification

Access control in this protocol is enforced through trust gates evaluated per-capability against peer classification. A trust gate is a predicate that takes a peer's resolved classification for a specific capability as input and produces ALLOW or DENY as output.

6.1 Built-in Tiers

Two built-in classification tiers are defined normatively:

UNRESTRICTED: The peer is allowed to activate all advertised capabilities with no scope constraints applied.

DENIED: The peer is not allowed to activate any capabilities. If all capabilities resolve to DENIED, the session MUST be closed.

All classification tiers between UNRESTRICTED and DENIED are implementation-defined. Their definition SHOULD be documented in a Peer Classification Profile as described in Appendix A.

6.2 Classification Authority

A peer's classification is determined by the receiving peer using local policy. A connecting peer MAY assert a classification in its capability declaration via the classification field as a hint or credential. The receiving peer MAY accept, override, or ignore this assertion.

Classification resolution MUST complete before capability activation. Because classification is per-capability, a single peer MAY be classified differently for different capabilities within the same session (e.g., UNRESTRICTED for core.session.heartbeat.1 but a restricted tier for core.data.stream.1).

6.3 Trust Gate Evaluation

For each capability in the received discovery manifest, the receiving peer MUST evaluate a trust gate using the resolved classification for that capability before the capability is considered for the active set.

If trust gate evaluation produces DENY for all capabilities, the session outcome MUST be set to NONE and the session MUST transition to CLOSED.

If trust gate evaluation produces DENY for a subset of capabilities, the denied capabilities MUST be excluded from the peer's computed active set. The session outcome depends on the intersection of the remaining capabilities with the remote peer's declarations.

If re-evaluation following a rebroadcast (§8) results in a previously ALLOW capability receiving DENY, that capability MUST be removed from the active set before the CONFIRM for the new exchange is sent.

7. Session Lifecycle

7.1 Session States

The following states govern the lifecycle of a peer relationship. PEER_VISIBLE and DENIED are pre-session dispositions; no transport connection exists. HANDSHAKE, CAPABILITY_EXCHANGE, and ACTIVE are session states. NONE is a terminal session outcome that always leads to CLOSED.

PEER_VISIBLE: Peer is known. No session is active. No protocol obligations.

HANDSHAKE: Mutual authentication is in progress. No capability exchange occurs in this state.

CAPABILITY_EXCHANGE: Peers are exchanging discovery manifests and confirming the active set. Entry point after HANDSHAKE. Re-entry point after any hash change

during ACTIVE.

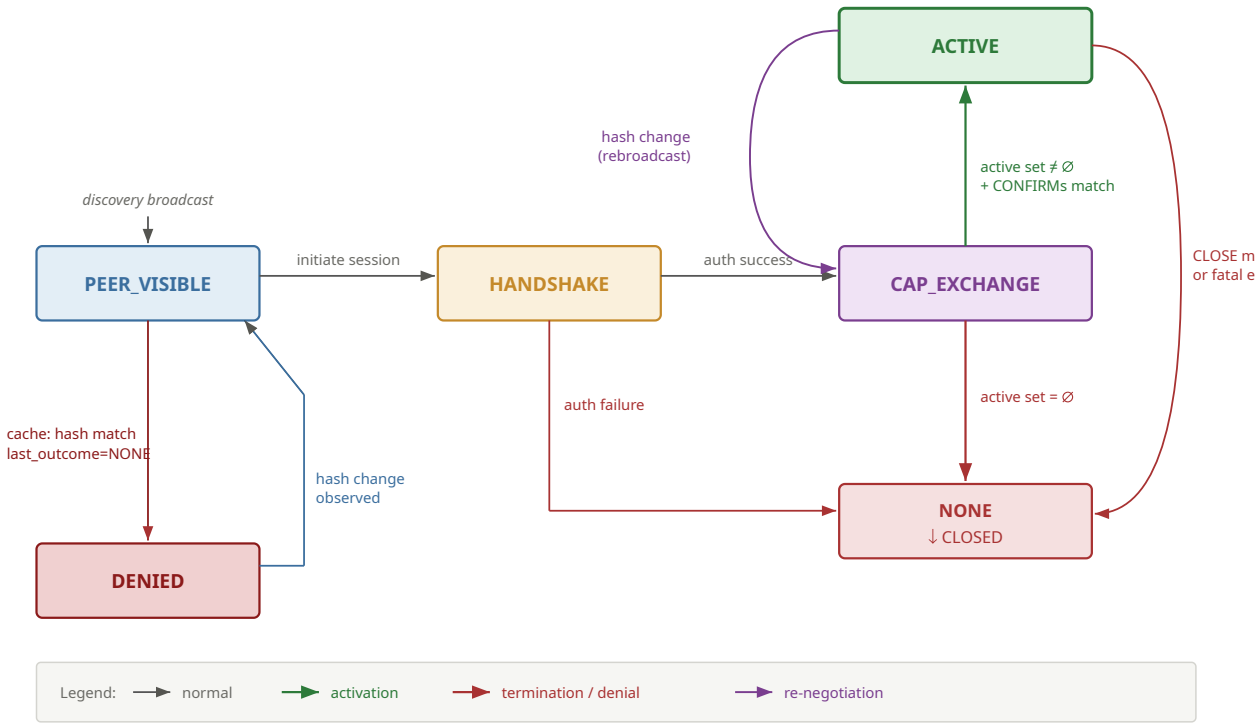
ACTIVE: At least one capability has been mutually negotiated, confirmed by both peers, and activated. Normal operating state.

NONE: No capabilities were negotiated or confirmed. Terminal outcome. Always results in session close.

DENIED: Pre-session disposition. Peer hash matches a cached NONE outcome. Negotiation is skipped. No transport connection is opened.

7.1.1 State Transition Diagram (Informational)

The following diagram is informational. The normative state transition table in §7.1.2 is authoritative where any discrepancy exists.



7.1.2 Normative State Transition Table

This table is the authoritative reference for session state transitions. Implementations MUST NOT allow transitions not listed here.

Current State	Event / Condition	Next State	Action(s)
PEER_VISIBLE	Local peer initiates connection	HANDSHAKE	Open transport connection. Begin mutual authentication per transport binding (Appendix C). If both peers simultaneously initiate, apply glare resolution (§7.1.3).
PEER_VISIBLE	Cache lookup: `last_hash == current_hash` AND `last_outcome == NONE`	DENIED	Record `outcome: DENIED` in peer cache. Do NOT open transport connection.
DENIED	New broadcast received with `hash ≠ last_hash`	PEER_VISIBLE	Update cache entry with new hash. Re-evaluate for session initiation.
HANDSHAKE	Mutual authentication succeeds	CAP_EXCHANGE	Send OFFER message (§5.3.3) with full discovery manifest. Await remote OFFER.
HANDSHAKE	Authentication failure or timeout	CLOSED	Send CLOSE with reason_code 2. Tear down transport. Peer reverts to PEER_VISIBLE .
CAP_EXCHANGE	Both OFFERs and both CONFIRMs received; reconciled active set ≠ ∅	ACTIVE	Activate the reconciled active set with reconciled scope parameters. Begin normal operation.
CAP_EXCHANGE	Both OFFERs received; local intersection = ∅	NONE → CLOSED	Send CONFIRM with empty active set. Send CLOSE with reason_code 1. Record `last_outcome: NONE` in peer cache. Tear down transport.
CAP_EXCHANGE	CONFIRMs received; reconciled active set = ∅	NONE → CLOSED	Send CLOSE with reason_code 1. Record `last_outcome: NONE` in peer cache. Tear down transport.
CAP_EXCHANGE	CONFIRM contains personal_hash ≠ previously	CAP_EXCHANGE	Remote state changed mid-exchange. Discard current exchange. Restart from OFFER. Increment local retry counter. If

Current State	Event / Condition	Next State	Action(s)
	received OFFER hash		retry limit exceeded, send CLOSE with reason_code 4 and transition to CLOSED.
`CAP_EXCHANGE`	Timeout: remote OFFER or CONFIRM not received within implementation-defined window	`CLOSED`	Send CLOSE with reason_code 4 if transport allows. Tear down transport. Do NOT cache as NONE (timeout is not a negotiation outcome).
`ACTIVE`	Remote peer rebroadcasts with new personal_hash	`CAP_EXCHANGE`	Re-enter capability exchange. Existing active set remains operational until new exchange completes (§8.4).
`ACTIVE`	Local capability state changes (any §8.1 trigger)	`CAP_EXCHANGE`	Recompute personal_hash, increment sequence_num, send new OFFER. Existing active set remains operational until re-exchange completes (§8.4).
`ACTIVE`	CLOSE message received from remote peer	`CLOSED`	Deactivate all capabilities. Tear down transport. Peer reverts to `PEER_VISIBLE` .
`ACTIVE`	Heartbeat failure (timeout exceeded)	`CAP_EXCHANGE`	Attempt re-exchange. If re-exchange produces NONE, transition to CLOSED.
`ACTIVE`	Ungraceful disconnect (transport error)	`CLOSED`	Deactivate all capabilities. Peer reverts to `PEER_VISIBLE` . Implementation MAY attempt reconnection.

7.1.3 Glare Resolution

If both peers simultaneously transition from PEER_VISIBLE to HANDSHAKE toward each other (simultaneous open), the peer with the lexicographically lower peer_id MUST abandon its outbound connection attempt and accept the inbound

connection. The peer with the lexicographically higher peer_id proceeds with its outbound connection. This produces exactly one session per peer pair.

7.2 Capability Exchange — Four-Message Sequence

The following is the normative message sequence for capability exchange. This sequence is used both for initial exchange after a successful handshake and for re-exchange triggered by a hash change during ACTIVE state. All messages use the CBOR wire format defined in §5.

```
-- Phase 1: Capability Advertisement --
1.  A -> B : OFFER   [ manifest_A ]              (§5.3.3)
2.  B -> A : OFFER   [ manifest_B ]              (§5.3.3)

-- Phase 2: Active Set Confirmation --
    Each peer independently computes intersection + trust gate evaluation.
    Each peer sends its computed active set and accepted parameters.

3.  A -> B : CONFIRM [ hash_A, active_set_A, ?accepted_params_A ]  (§5.3.4)
4.  B -> A : CONFIRM [ hash_B, active_set_B, ?accepted_params_B ]  (§5.3.4)

-- Reconciliation --
    Each peer computes: final_set = active_set_A ∩ active_set_B
    For each capability in final_set:
        final_params = min(accepted_params_A, accepted_params_B)
    If final_set ≠ ∅  → both enter ACTIVE with final_set and final_params
    If final_set = ∅   → both enter NONE → CLOSED
```

Each peer MUST send its full discovery manifest in the OFFER message. Partial or delta offers are not permitted.

The active_set in the CONFIRM message is the sorted list of capability names that the sending peer considers activated after intersection computation and trust gate evaluation. Capabilities MUST be sorted lexicographically by name.

The personal_hash in each CONFIRM MUST match the personal_hash from that peer's preceding OFFER. If a peer receives a CONFIRM whose personal_hash differs from the hash in the OFFER previously received from that peer, the peer's state changed during the exchange. The receiving peer MUST discard the current exchange and restart from step 1. Implementations SHOULD enforce a retry limit; exceeding the retry limit MUST result in transition to CLOSED.

The reconciled active set (final_set) is the intersection of both CONFIRM active sets. If the reconciled set is empty, the session outcome MUST be NONE and the session MUST transition to CLOSED. If the reconciled set is non-empty, both peers enter ACTIVE with the reconciled set as the definitive active capabilities and the reconciled parameters as the definitive scope constraints.

7.2.1 Alternate Paths

Empty intersection at Phase 1: If a peer computes an empty intersection after receiving the remote OFFER (before sending CONFIRM), it MUST still send a CONFIRM with an empty active set. This allows the remote peer to distinguish "no match" from "timeout."

Hash change during exchange: Handled by the hash consistency check in the CONFIRM phase. See §7.2 above.

Timeout: If a peer does not receive the remote OFFER within an implementation-defined timeout, it MUST transition to CLOSED. If a peer does not receive the remote CONFIRM within an implementation-defined timeout after sending its own CONFIRM, it MUST transition to CLOSED. Timeouts MUST NOT be cached as NONE outcomes.

Simultaneous OFFER: Both peers may send OFFER before receiving the other's OFFER. This is the expected case and does not require special handling; the protocol is designed for concurrent OFFER transmission.

Protocol version mismatch: If a peer receives an OFFER with a protocol_version it does not support, it MUST send a CLOSE with reason_code 3 (version_unsupported) and transition to CLOSED.

7.3 Parameter Negotiation

Scope parameter negotiation occurs within the CONFIRM phase of the four-message exchange. Each peer advertises its offered parameters in the OFFER (via the scope field of each capability declaration). Each peer then declares its accepted parameters in the CONFIRM (via the optional accepted_params field).

Reconciliation follows a most-restrictive-wins rule for numeric parameters: the final value for rate_limit is the minimum of both peers' accepted values; the final value for ttl is the minimum of both peers' accepted values. This ensures that neither peer is committed to more than it offered.

For non-numeric parameters (e.g., codec, encoding), reconciliation rules MUST be defined by the capability's specification (Appendix B). If a capability's specification does not define reconciliation for a non-numeric parameter, the PROVIDE peer's value takes precedence.

If a peer determines that the reconciled parameters are unacceptable (e.g., the reconciled bitrate is below a usable threshold), it MAY exclude that capability from its active set in the CONFIRM, which will cause the capability to be absent from the reconciled final set.

7.4 Role-Based Intersection

Intersection is computed per capability using the role fields of both peers' declarations. The following rules define a match:

PROVIDE + CONSUME: Match. The PROVIDE peer acts as server; the CONSUME peer acts as client.

CONSUME + PROVIDE: Match. Identical to above from opposite perspective.

BOTH + PROVIDE: Match. The BOTH peer acts as client; the PROVIDE peer acts as server.

BOTH + CONSUME: Match. The BOTH peer acts as server; the CONSUME peer acts as client.

BOTH + BOTH: Match ONLY if both declarations have mutual: true. Both peers participate as simultaneous server and client. If either declaration has mutual: false or the field is absent, no match.

PROVIDE + PROVIDE: No match.

CONSUME + CONSUME: No match.

7.5 Connection Archetypes

Three connection archetypes are defined. Worked message sequence examples for each are provided in Appendix E.

NONE / NONE: Neither peer offers any capability the other can consume. Session outcome is NONE. Session closes.

ASYMMETRIC: One peer provides capabilities the other consumes. Session outcome is ACTIVE.

MUTUAL: Both peers provide capabilities to each other simultaneously. Both declare BOTH (with mutual: true) or complementary PROVIDE/CONSUME pairs. Session outcome is ACTIVE.

7.6 Session Termination

A session outcome of NONE MUST result in session close. Implementations MUST send a CLOSE message (§5.3.5) with the appropriate reason_code, tear down the transport connection, and record the outcome in the peer cache.

A closing peer SHOULD send a CLOSE message carrying its current personal hash and reason_code 0 (normal) before terminating the connection. Receipt of a CLOSE message MUST cause the receiving peer to deactivate all capabilities and tear down the transport connection.

Ungraceful disconnection (no CLOSE message received) SHOULD be detected by the heartbeat mechanism (core.session.heartbeat.1, Appendix B.1). A heartbeat failure within the declared timeout constitutes a session failure and MUST trigger CAPABILITY_EXCHANGE re-entry. If re-exchange produces NONE, the session closes.

7.7 Capability Activation Sequence

When a session transitions to ACTIVE with a non-empty reconciled active set, capabilities in the active set MAY define an activation exchange — a bounded sequence of messages that MUST complete before the capability is considered fully operational.

Activation exchanges, if defined, proceed in the order of the capability names sorted lexicographically (the same order used in the CONFIRM active set). Each capability's activation exchange MUST complete (or time out) before the next capability's activation exchange begins.

A capability that defines no activation exchange is immediately operational upon entering the active set.

A capability whose activation exchange fails or times out MUST be removed from the active set. If removal leaves the active set empty, the session MUST transition to

NONE.

Activation exchanges MUST be idempotent: if a re-exchange (§8) produces an active set containing a capability that was already active and whose activation exchange previously succeeded, the activation exchange MUST NOT be repeated.

8. Change Detection and Rebroadcast

8.1 Rebroadcast Triggers

Any change to a peer's capability state MUST cause that peer to recompute its personal hash and rebroadcast its full discovery manifest. The following changes are normative triggers:

Capability addition: A new capability is added to the manifest. MUST rebroadcast.

Capability removal: An existing capability is removed. MUST rebroadcast.

Scope parameter change: Any scope parameter changes for any capability. MUST rebroadcast.

Classification change: The peer's own group membership changes. MUST rebroadcast immediately. All capabilities granted under the prior classification MUST be re-evaluated before the CONFIRM for the new exchange is sent.

Role change: A capability's role (PROVIDE/CONSUME/BOTH) changes. MUST rebroadcast.

8.2 Rebroadcast Behavior

Rebroadcast MUST transmit the full discovery manifest. Delta updates are not permitted in this version of the protocol.

The sequence_num field MUST be incremented before rebroadcast.

A peer that receives a discovery manifest whose personal hash differs from the last stored hash for that peer MUST re-enter CAPABILITY_EXCHANGE state.

8.3 Advisory Change Flags

A rebroadcast MAY include an advisory change_flags field in the OFFER message (§5.3.3). This field is non-normative; receiving peers MUST still perform full re-evaluation regardless of its contents. Integer values for change flags are defined in the change-flag enumeration in §5.3.3. Implementations MAY use change_flags to optimize re-evaluation (e.g., fast-pathing scope-only changes) but MUST NOT rely on them for correctness.

8.4 Active Set Continuity During Re-exchange

When a session transitions from ACTIVE to CAPABILITY_EXCHANGE due to a hash change, the existing active set MUST remain operational until the new exchange completes. Capabilities in the current active set continue to function during re-negotiation. Once the new exchange produces a reconciled active set (§7.2), the new set replaces the prior set atomically. Capabilities present in the prior set but absent from the new set MUST be deactivated. Capabilities present in both sets MUST NOT be interrupted.

8.5 Re-evaluation Obligations

On receipt of a rebroadcast, the receiving peer MUST re-evaluate classification and scope for all capabilities in the new manifest.

If re-evaluation results in revocation of a previously ALLOW capability, revocation MUST be applied before the CONFIRM for the new exchange is sent.

9. Auto-Deny and Peer Cache

9.1 Peer Cache Structure

Each peer maintains a local peer cache. The cache records the outcome of past interactions indexed by peer identity and capability hash.

```
peer_cache_entry {
```

```
    peer_id          : bstr      ; cryptographic identity handle
    last_hash        : bstr      ; personal hash from last exchange
    last_outcome     : uint      ; ACTIVE=1 | NONE=2 | DENIED=3
    last_seen        : uint      ; timestamp (seconds since epoch)
}
```

Implementations that support auto-deny MUST store peer_id, last_hash, last_outcome, and last_seen for every session that terminates with outcome NONE. The last_seen field MUST be updated on every interaction with the peer, including discovery broadcasts.

The peer cache MUST NOT be transmitted to third parties.

The peer cache SHOULD be treated as sensitive local state and protected with equivalent controls to authentication credentials.

9.2 Auto-Deny Behavior

Auto-deny is RECOMMENDED. Implementations that support auto-deny MUST implement the following behavior: if a peer is PEER_VISIBLE and its current personal hash matches last_hash in the cache and last_outcome is NONE, the implementation MUST record outcome DENIED and MUST NOT initiate a handshake with that peer.

If auto-deny is implemented, the cache hash check MUST precede any handshake attempt.

If a peer's personal hash changes relative to last_hash in the cache, the implementation SHOULD attempt a new handshake regardless of last_outcome.

9.3 Reconnection on Capability Change

A peer whose manifest changes from containing no capabilities to containing one or more capabilities SHOULD attempt reconnection to known PEER_VISIBLE peers. A peer that adds capabilities to a non-empty manifest SHOULD rebroadcast to all active sessions per §8 and MAY attempt connection to PEER_VISIBLE peers whose cached last_outcome was NONE (since the hash has changed, auto-deny will not apply).

10. Security Considerations

10.1 Assumed Precondition

This protocol assumes a mutually authenticated session is established before capability exchange begins. Both peers MUST have verified each other's cryptographic identity handle prior to entering CAPABILITY_EXCHANGE state. The mechanism of mutual authentication is not defined here; implementations SHOULD use an established protocol such as TLS 1.3 mutual auth, the Noise Protocol Framework, or equivalent.

10.2 Replay Prevention

The sequence_num field in the discovery manifest is a monotonically increasing integer included in the personal hash computation. A receiver that observes a sequence_num not greater than the last seen value from that peer MUST discard the manifest as a potential replay.

10.3 Discovery Broadcast Exposure

When a full discovery manifest is included in a broadcast, the peer's capability set is visible to all receivers, including unauthenticated observers. This is a deliberate design trade-off enabling pre-session compatibility filtering. Implementations that require capability confidentiality prior to authentication MUST use lightweight discovery (§3.1) and restrict discovery to transports that provide channel-level encryption.

10.4 Peer Cache Privacy

The peer cache contains a record of observed peer identities, their capability hashes, and session outcomes. This constitutes a potentially sensitive log of peer interactions. Implementations MUST NOT transmit this data to third parties. Implementations SHOULD apply access controls to the cache equivalent to those applied to private key material.

10.5 Sybil Resistance

Sybil resistance is an identity-layer concern and is out of scope for this protocol. By requiring mutually authenticated sessions as a precondition, this protocol delegates

Sybil resistance to the identity layer. Implementations SHOULD select identity mechanisms with appropriate Sybil resistance properties for their deployment context.

10.6 Active Set Confirmation and Trust Asymmetry

The four-message exchange (§7.2) ensures that both peers explicitly confirm the active set before capability activation. However, because each peer applies its own local classification policy, the two peers may arrive at different active sets from the same pair of manifests. The CONFIRM reconciliation step resolves this by taking the intersection of both active sets, ensuring that no capability is activated unless both peers independently approved it. Implementations MUST NOT activate a capability that was not present in both CONFIRM messages.

10.7 Wire Format Security

CBOR decoders MUST enforce limits on nesting depth, map key count, and total message size to prevent resource exhaustion attacks. Implementations SHOULD reject messages exceeding implementation-defined size limits before full decoding. The deterministic encoding profile (§5.2) mitigates certain classes of canonicalization attacks that could cause hash mismatches between conforming implementations.

COSE-signed credentials (§5.4) MUST be verified before the claims within them are used for classification. Implementations MUST NOT trust the contents of a COSE structure whose signature or MAC verification has failed.

10.8 Limits of Protocol-Level Trust

This protocol provides mechanisms for peers to exchange identity, capability, and build information. It does not and cannot guarantee that a remote peer is behaving honestly.

A peer that controls its own runtime environment can misrepresent any self-reported value, including the personal hash, capability declarations, scope parameters, build attestation data, and classification credentials. No software-only protocol can verify the runtime behavior of a remote process without a hardware trust root.

The trust gates, classification system, and build attestation capability defined in this specification raise the cost of undetected misbehavior. They do not eliminate it. Implementations SHOULD NOT treat protocol-level trust mechanisms as sufficient for high-stakes authorization decisions without additional out-of-band verification.

Peer operators are responsible for evaluating the trustworthiness of the peers they connect to through means appropriate to their deployment context.

11. IANA Considerations

This document requests the following IANA registries:

P2P-CAPDISC Well-Known Capabilities Registry — maps capability name strings under the reserved "core.*" namespace to specification documents. Registration requires Standards Action.

P2P-CAPDISC Hash Algorithm Registry — maps algorithm identifier strings to hash algorithm specifications. Entries SHOULD reference algorithms registered in the IANA Named Information Hash Algorithm Registry.

P2P-CAPDISC Scope Parameter Registry — maps scope parameter integer keys to field names, type definitions, units, and applicability constraints. Keys 1 (rate_limit) and 2 (ttl) are registered by this document. Scope parameter keys 1–15 are reserved for parameters defined by this specification and its normative extensions. Keys 16–127 are available for capability-specific parameters registered in the P2P-CAPDISC Scope Parameter Registry. Keys 128+ are available for application-defined parameters without registration.

P2P-CAPDISC Message Type Registry — maps message type integer values to message names and specification sections. Values 1 (OFFER), 2 (CONFIRM), and 3 (CLOSE) are registered by this document. See §5.3.6 for the message type allocation policy.

P2P-CAPDISC Close Reason Code Registry — maps reason code integer values to reason names. Values 0–4 and 255 are registered by this document.

12. References

12.1 Normative References

RFC 2119 — Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, March 1997.

RFC 8446 — Rescorla, E., "The Transport Layer Security (TLS) Protocol Version 1.3", August 2018.

STD 94 / RFC 8949 — Bormann, C. and P. Hoffman, "Concise Binary Object Representation (CBOR)", December 2020.

RFC 8610 — Birkholz, H., Vigano, C., and C. Bormann, "Concise Data Definition Language (CDDL): A Notational Convention to Express Concise Binary Object Representation (CBOR) and JSON Data Structures", June 2019.

STD 96 / RFC 9052 — Schaad, J., "CBOR Object Signing and Encryption (COSE): Structures and Process", August 2022.

RFC 9053 — Schaad, J., "CBOR Object Signing and Encryption (COSE): Initial Algorithms", August 2022.

12.2 Informative References

RFC 8259 — Bray, T., Ed., "The JavaScript Object Notation (JSON) Data Interchange Format", STD 90, December 2017.

RFC 8392 — Jones, M., Wahlstroem, E., Erdtman, S., and H. Tschofenig, "CBOR Web Token (CWT)", May 2018.

RFC 8742 — Bormann, C., "Concise Binary Object Representation (CBOR) Sequences", February 2020.

RFC 9054 — Schaad, J., "CBOR Object Signing and Encryption (COSE): Hash Algorithms", August 2022.

Noise Protocol Framework — Perrin, T., "The Noise Protocol Framework", http://noiseprotocol.org/noise.html, July 2018.

W3C Verifiable Credentials Data Model — Sporny, M., et al., https://www.w3.org/TR/vc-data-model/

libp2p Identify Protocol — https://github.com/libp2p/specs/tree/master/identify

Appendix A. Trust Credential Formats

This appendix defines how common credential formats map to the classification field in the capability declaration. These mappings are informational. Implementations MAY support any subset. COSE-based credentials (§5.4) are RECOMMENDED for new deployments.

A.1 CBOR Web Token (CWT) via COSE

A CWT (RFC 8392) is the CBOR equivalent of a JWT and is the RECOMMENDED credential format for this protocol. The CWT is encoded as a COSE_Sign1 structure and placed in the classification field as a credential_blob (byte string). The receiving peer MUST verify the COSE signature before using the claims. The group reference is extracted from a designated claim. The CWT MUST include an exp claim; expired tokens MUST be rejected.

A.2 JSON Web Token (JWT)

A JWT may be placed in the classification field as a credential_blob. The JWT is encoded as a UTF-8 byte string within the CBOR message. The receiving peer MUST verify the JWT signature against a trusted public key before using the claims. Implementations SHOULD prefer CWT (A.1) over JWT for new deployments to avoid format conversion overhead.

A.3 W3C Verifiable Credentials

A W3C Verifiable Credential may be placed in the classification field. The receiving peer MUST verify the credential proof before accepting the credential subject claims as the group reference. This approach supports decentralized identity scenarios without a central credential issuer.

A.4 Local Policy Reference

The classification field MAY carry a group_ref text string that the receiving peer resolves against a local policy store. This is the simplest form of classification and requires no cryptographic verification of the field itself. The security of this approach depends entirely on the integrity of the authenticated session — the peer_id is trusted, and the local policy maps peer_id to group.

A.5 Macaroons

Macaroons may be used as credential_blobs where capability-scoped, delegatable authorization is required. The caveats in the macaroon encode the scope constraints

directly, allowing the receiving peer to verify that the presented token authorizes specifically the capabilities being requested.

Appendix B. Core Capability Catalog

This appendix defines the core capability implementations using the namespace grammar defined in §4.4, the capability definition requirements of §4.6, and the capability message envelope of §5.3.7. These definitions are normative for implementations that claim to support the named capabilities.

The catalog is organized into three tiers based on the network layer the capability addresses: session (§B.1–B.4), network (§B.5–B.7), and data (§B.8–B.11).

B.1 core.session.heartbeat.1

Name	`core.session.heartbeat.1`
Role	BOTH (mutual: true)
Purpose	Session liveness detection
Params	`interval_ms: uint` , `timeout_ms: uint`
Scope	`ttl` applies; `rate_limit` does not apply
Message types	PING (4), PONG (5)
Conformance	**Mandatory** (§D.1)

```
ping-message = { 1 : 4, 2 : uint }   ; message_type=4, timestamp
pong-message = { 1 : 5, 2 : uint }   ; message_type=5, echo timestamp
```

Both peers MUST declare role BOTH with mutual: true for this capability to activate. Both peers send PING at interval_ms. Failure to receive PONG within timeout_ms constitutes session failure and MUST trigger CAPABILITY_EXCHANGE re-entry. This is the foundation that every other capability relies on — without liveness, no other guarantee can be reasoned about.

Negotiation: interval_ms uses most-restrictive-wins (higher value = less frequent). timeout_ms uses most-restrictive-wins (lower value = stricter).

B.2 core.session.attest.1

Name	`core.session.attest.1`
Role	BOTH (mutual: true)
Purpose	Mutual exchange of build identity at session activation
Params	See `attest-params` below
Scope	`ttl` applies; `rate_limit` does not apply
Message types	BUILD_ATTEST (6)
Activation exchange	Yes — single message each direction upon entering active set
Conformance	**Recommended**

```
build-attest = { 1 : 6,
    2 : attest-params
}

attest-params = {
    1 : uint,          ; spec_version — protocol spec version this build targets
    2 : tstr,          ; lib_name — implementation identifier (e.g. "p2pcd-core")
    3 : tstr,          ; lib_version — semantic version of the library
    4 : tstr,          ; source_repo — canonical source URL
    5 : bstr,          ; source_hash — commit hash the build was produced from
    6 : bstr,          ; binary_hash — hash of the running binary image
    7 : tstr,          ; hash_algorithm — IANA name for binary_hash (e.g. "sha-25
    ? 8 : tstr,        ; build_target — compiler target triple
    ? 9 : tstr,        ; build_profile — "release" | "debug" | custom
    ? 10 : bstr,       ; signature — detached signature over canonical CBOR of fi
    ? 11 : bstr,       ; signer_id — public key of the signer
    ? 12 : [* patch-decl]  ; declared modifications to source
}

patch-decl = {
    1 : tstr,          ; patch_name — human-readable label
    2 : bstr           ; patch_hash — hash of the patch content
}
```

When core.session.attest.1 enters the active set after CONFIRM reconciliation, both peers immediately send a single build-attest message (§7.7). There is no challenge, no ongoing re-verification, no response expected.

Each peer evaluates the received attestation against local policy and updates the peer's classification accordingly. The evaluation is entirely local — the protocol does not define pass/fail semantics.

What this verifies: Spec compatibility (spec_version). Implementation identity (lib_name, lib_version). Source provenance (source_repo, source_hash). Binary integrity (binary_hash against published reproducible builds). Build authority (signature, signer_id). Declared modifications (patch-decl).

What this does NOT verify: Runtime honesty — a peer that controls its own process can forge any self-reported value. Behavioral conformance — a correct binary on a compromised OS may behave non-conformantly. Shared library integrity — only the main executable is hashed.

Threat	Detected?	Mechanism
Trojanized binary from compromised mirror	**Yes**	binary_hash mismatch against published release
Accidental dirty build on production mesh	**Yes**	source_hash doesn't match a release commit
Unknown fork joining the mesh	**Yes**	source_repo points to unrecognized repository
Patched build with undeclared modifications	**Partial**	binary_hash mismatch, but no patch-decl
Determined adversary forging all fields	**No**	Self-reported values; no hardware trust root
Runtime code injection (LD_PRELOAD, ptrace)	**No**	Binary on disk is correct; runtime is not
Compromised shared libraries	**No**	Only main executable is hashed

B.3 core.session.timesync.1

Name	`core.session.timesync.1`
Role	BOTH (mutual: true)
Purpose	Clock offset estimation between peers
Params	`precision_ms: uint` — desired precision floor

Scope	`rate_limit` applies (sync probes/sec); `ttl` applies
Message types	TIME_REQ (7), TIME_RESP (8)
Conformance	Recommended

```
time-req  = { 1 : 7,  2 : uint }   ; message_type=7, t1 (sender timestamp ms)
time-resp = { 1 : 8,  2 : uint,    ; message_type=8, t1 (echo)
                       3 : uint,    ; t2 (receiver timestamp at receipt)
                       4 : uint }   ; t3 (receiver timestamp at send)
```

NTP-style four-timestamp exchange. The initiator records t4 on receipt. Offset = ((t2 - t1) + (t3 - t4)) / 2. RTT = (t4 - t1) - (t3 - t2). Coordinated timestamps are required for event ordering, TTL enforcement across peers, certificate validation, log correlation, and replay detection.

Negotiation: precision_ms uses most-restrictive-wins (higher value = less precise = less traffic).

B.4 core.session.latency.1

Name	`core.session.latency.1`
Role	BOTH (mutual: true)
Purpose	One-way and round-trip latency measurement
Params	`sample_interval_ms: uint , window_size: uint`
Scope	`rate_limit` applies; `ttl` applies
Message types	LAT_PING (9), LAT_PONG (10)
Conformance	Optional

```
lat-ping = { 1 : 9,  2 : uint,   ; message_type=9, sequence
                      3 : uint }  ; t_send (sender timestamp μs)
lat-pong = { 1 : 10, 2 : uint,   ; message_type=10, sequence (echo)
                      3 : uint,   ; t_send (echo)
                      4 : uint }  ; t_recv (responder timestamp μs)
```

Distinct from heartbeat. Heartbeat answers "are you alive?" with a binary yes/no. Latency ping answers "how far away are you?" with a measurement. The window_size param controls how many samples to retain for rolling statistics (mean, p50, p99). Latency-aware peer selection is fundamental to relay routing, stream quality adaptation, and any application that needs to choose the best peer from a set.

Negotiation: sample_interval_ms uses most-restrictive-wins (higher = less frequent). window_size uses most-restrictive-wins (lower = less memory).

B.5 core.network.endpoint.1

Name	`core.network.endpoint.1`
Role	PROVIDE | CONSUME
Purpose	Peer reports the requesting peer's externally observed network address
Params	`include_geo: bool` — whether to include approximate geolocation
Scope	`rate_limit` applies; `ttl` applies
Message types	WHOAMI_REQ (11), WHOAMI_RESP (12)
Conformance	Recommended

```
whoami-req  = { 1 : 11 }
whoami-resp = { 1 : 12,
    2 : tstr,          ; observed_addr (e.g. "203.0.113.42:41312")
    3 : uint,          ; addr_family (4 = IPv4, 6 = IPv6)
    ? 4 : tstr,        ; observed_hostname (reverse DNS, if available)
    ? 5 : [float, float]  ; approx_geo [lat, lon] (if include_geo)
}
```

The P2P-CD equivalent of STUN. The PROVIDE peer looks at the transport-level source address of the authenticated session and reports it back. The CONSUME peer learns its public-facing address from the perspective of this specific peer. NAT traversal, multi-homing, and peer advertisement all require a node to know its own externally visible address.

Negotiation: include_geo is provider-takes-precedence (the reflector decides whether to provide geolocation).

B.6 core.network.relay.1

Name	`core.network.relay.1`
Role	PROVIDE | CONSUME
Purpose	Third-party relay — forwarding framed messages between two peers that cannot directly connect
Params	`max_circuits: uint` , `max_bandwidth_kbps: uint` , `relay_ttl: uint`
Scope	`rate_limit` applies (circuit creation rate); `ttl` applies (circuit lifetime)
Message types	CIRCUIT_OPEN (13), CIRCUIT_DATA (14), CIRCUIT_CLOSE (15)
Conformance	Recommended

```
circuit-open = { 1 : 13,
    2 : bstr,          ; target_peer_id
    3 : uint,          ; requested_bandwidth_kbps
    ? 4 : uint         ; requested_ttl (seconds)
}
circuit-data = { 1 : 14,
    2 : uint,          ; circuit_id (assigned by relay)
    3 : bstr           ; payload (opaque, end-to-end encrypted by peers)
}
circuit-close = { 1 : 15,
    2 : uint,          ; circuit_id
    ? 3 : uint         ; reason (0=normal, 1=target_unreachable,
                       ;         2=bandwidth_exceeded, 3=ttl_expired)
}
```

A PROVIDE peer acts as a public relay. A CONSUME peer requests a circuit to a target peer that it cannot reach directly. The relay forwards opaque frames; it never inspects payload content.

The relay MUST NOT have access to plaintext content. All circuit-data payloads are end-to-end encrypted between the two endpoint peers using their own session keys. The relay only sees ciphertext. This is a MUST-level requirement.

Negotiation: max_bandwidth_kbps uses most-restrictive-wins. max_circuits is provider-takes-precedence (the relay decides its own capacity). relay_ttl uses most-restrictive-wins.

B.7 core.network.peerexchange.1

Name	`core.network.peerexchange.1`
Role	BOTH (mutual: true)
Purpose	Gossip-style exchange of known peer addresses and capability hashes
Params	`max_peers: uint , include_capabilities: bool`
Scope	`rate_limit` applies (exchanges/sec); `ttl` applies
Message types	PEX_REQ (16), PEX_RESP (17)
Conformance	Recommended

```
pex-req = { 1 : 16,
    ? 2 : tstr,        ; filter_capability (only return peers offering this)
    ? 3 : uint         ; max_results
}
pex-resp = { 1 : 17,
    2 : [* peer-info]
}
peer-info = {
    1 : bstr,          ; peer_id
    2 : bstr,          ; personal_hash
    3 : tstr,          ; reachable_addr
    4 : uint,          ; last_seen (seconds since epoch)
    ? 5 : [* tstr]     ; capability_names (if include_capabilities)
}
```

Peer Exchange is how sparse networks bootstrap into connected meshes without relying on a central rendezvous server. Peers share their knowledge of other reachable peers, optionally filtered by capability. The personal_hash is included so the receiver can apply auto-deny checks before attempting connection.

Security: A peer MUST NOT include peers in a PEX response that have not consented to peer exchange (i.e., that are not themselves advertising

core.network.peerexchange.1).

Negotiation: max_peers uses most-restrictive-wins. include_capabilities is provider-takes-precedence.

B.8 core.data.blob.1

Name	`core.data.blob.1`
Role	PROVIDE | CONSUME
Purpose	Reliable transfer of a finite, integrity-verified blob
Params	`max_blob_bytes: uint` , `chunk_size: uint` , `hash_algorithm: tstr`
Scope	`rate_limit` applies (bytes/sec); `ttl` applies
Message types	BLOB_REQ (18), BLOB_OFFER (19), BLOB_CHUNK (20), BLOB_ACK (21)
Conformance	Recommended

```
blob-req = { 1 : 18,
    2 : bstr,          ; content_hash (identifies the blob)
    3 : tstr,          ; hash_algorithm
    ? 4 : uint         ; offset (resume from byte N)
}
blob-offer = { 1 : 19,
    2 : bstr,          ; content_hash (echo)
    3 : uint,          ; total_size (bytes)
    4 : uint,          ; chunk_size (bytes)
    ? 5 : uint         ; status (0=available, 1=not_found, 2=too_large)
}
blob-chunk = { 1 : 20,
    2 : bstr,          ; content_hash
    3 : uint,          ; offset
    4 : bstr           ; data
}
blob-ack = { 1 : 21,
    2 : bstr,          ; content_hash
    3 : uint,          ; bytes_received
    ? 4 : uint         ; status (0=continue, 1=complete, 2=hash_mismatch, 3=abort
}
```

Content-addressed blob transfer with chunking, resume, and integrity verification. The blob is identified by its hash, not by a filename or path. The consumer requests a blob by hash; the provider streams chunks; the consumer verifies the hash over the complete reassembled blob.

The offset field in blob-req enables resume after transport interruption. The provider starts sending from that offset.

Negotiation: max_blob_bytes is provider-takes-precedence (the provider knows its storage). chunk_size uses most-restrictive-wins. hash_algorithm: if peers disagree, the provider's value takes precedence per §7.3.

B.9 core.data.rpc.1

Name	`core.data.rpc.1`
Role	PROVIDE | CONSUME
Purpose	Generic request/response RPC over the P2P-CD session
Params	`max_request_bytes: uint` , `max_response_bytes: uint` , `methods: [tstr]`
Scope	`rate_limit` applies; `ttl` applies
Message types	RPC_REQ (22), RPC_RESP (23)
Conformance	Optional

```
rpc-req = { 1 : 22,
    2 : uint,          ; request_id
    3 : tstr,          ; method
    4 : bstr           ; payload (CBOR-encoded, method-specific)
}
rpc-resp = { 1 : 23,
    2 : uint,          ; request_id (echo)
    3 : uint,          ; status (0=ok, 1=not_found, 2=error, 3=timeout)
    ? 4 : bstr         ; payload (CBOR-encoded, method-specific)
    ? 5 : tstr         ; error_message
}
```

Generic request/response semantics over an established P2P-CD session. The methods param in the capability declaration advertises which RPC methods the provider supports. Method semantics are defined by the application; the core capability only standardizes the envelope.

Negotiation: max_request_bytes and max_response_bytes use most-restrictive-wins. methods: the activated method set is the intersection of both peers' declared methods.

B.10 core.data.event.1

Name	`core.data.event.1`
Role	PROVIDE | CONSUME | BOTH
Purpose	Pub/sub event notification with topic filtering
Params	`topics: [tstr]` , `max_payload_bytes: uint`
Scope	`rate_limit` applies (events/sec); `ttl` applies
Message types	EVENT_SUB (24), EVENT_UNSUB (25), EVENT_MSG (26)
Conformance	Optional

```
event-sub = { 1 : 24,
    2 : [* tstr]       ; topic_filters (prefix match)
}
event-unsub = { 1 : 25,
    2 : [* tstr]       ; topic_filters
}
event-msg = { 1 : 26,
    2 : tstr,          ; topic
    3 : bstr,          ; payload (opaque, application-defined)
    4 : uint,          ; timestamp_ms
    ? 5 : bstr         ; source_peer_id (originator, if relayed)
}
```

Topic-based pub/sub over an active session. A PROVIDE peer accepts subscriptions and pushes matching events. A CONSUME peer subscribes to topics and receives events. BOTH with mutual: true enables bidirectional event exchange.

Topic filtering: topic_filters use prefix matching. Subscribing to ["chat."] receives chat.general, chat.random, etc. Subscribing to [""] (empty string prefix) receives everything. The topics param in the capability declaration advertises available topic prefixes so consumers can evaluate compatibility at OFFER time.

Negotiation: max_payload_bytes uses most-restrictive-wins. topics: the activated topic set is the intersection of provider's advertised topics and consumer's requested

filters.

B.11 core.data.stream.1

Name	`core.data.stream.1`
Role	PROVIDE | CONSUME
Purpose	Unidirectional continuous data stream
Params	`bitrate_kbps: uint , codec: tstr`
Scope	`ttl` applies; `rate_limit` does not apply (continuous stream)
Transport	Requires transport.udp or transport.quic
Conformance	Optional

Negotiation: bitrate_kbps uses most-restrictive-wins (min of provider and consumer). codec uses provider-takes-precedence per §7.3 fallback for non-numeric parameters.

B.12 Message Type Allocation Summary

Type	Message	Capability	Status
1	OFFER	(protocol)	Normative §5.3.3
2	CONFIRM	(protocol)	Normative §5.3.4
3	CLOSE	(protocol)	Normative §5.3.5
4	PING	`core.session.heartbeat.1`	Normative B.1
5	PONG	`core.session.heartbeat.1`	Normative B.1
6	BUILD_ATTEST	`core.session.attest.1`	Normative B.2
7	TIME_REQ	`core.session.timesync.1`	Normative B.3
8	TIME_RESP	`core.session.timesync.1`	Normative B.3
9	LAT_PING	`core.session.latency.1`	Normative B.4
10	LAT_PONG	`core.session.latency.1`	Normative B.4
11	WHOAMI_REQ	`core.network.endpoint.1`	Normative B.5

Type	Message	Capability	Status
12	WHOAMI_RESP	`core.network.endpoint.1`	Normative B.5
13	CIRCUIT_OPEN	`core.network.relay.1`	Normative B.6
14	CIRCUIT_DATA	`core.network.relay.1`	Normative B.6
15	CIRCUIT_CLOSE	`core.network.relay.1`	Normative B.6
16	PEX_REQ	`core.network.peerexchange.1`	Normative B.7
17	PEX_RESP	`core.network.peerexchange.1`	Normative B.7
18	BLOB_REQ	`core.data.blob.1`	Normative B.8
19	BLOB_OFFER	`core.data.blob.1`	Normative B.8
20	BLOB_CHUNK	`core.data.blob.1`	Normative B.8
21	BLOB_ACK	`core.data.blob.1`	Normative B.8
22	RPC_REQ	`core.data.rpc.1`	Normative B.9
23	RPC_RESP	`core.data.rpc.1`	Normative B.9
24	EVENT_SUB	`core.data.event.1`	Normative B.10
25	EVENT_UNSUB	`core.data.event.1`	Normative B.10
26	EVENT_MSG	`core.data.event.1`	Normative B.10
27	STREAM_OPEN	`core.data.stream.1`	Normative B.11
28	STREAM_DATA	`core.data.stream.1`	Normative B.11
29	STREAM_CLOSE	`core.data.stream.1`	Normative B.11
30	STREAM_CONTROL	`core.data.stream.1`	Normative B.11
31-35	(reserved)	—	Reserved for v2 core extensions
36+	(application)	(application-defined)	Open

Appendix C. Transport Bindings

This appendix defines how the capability exchange message sequence maps onto common transport protocols. Each binding is informational; implementations MAY define additional bindings. All messages use the CBOR wire format (§5) with 4-byte length-prefix framing over stream transports.

C.1 TCP

Discovery manifests are framed as length-prefixed CBOR messages over a TCP stream. Mutual authentication SHOULD be implemented using TLS 1.3 with client certificates. The HANDSHAKE state corresponds to the TLS handshake phase. CAPABILITY_EXCHANGE begins after the TLS handshake completes successfully.

C.2 QUIC

Discovery manifests are sent as length-prefixed CBOR messages on a dedicated QUIC stream (stream ID 0 by convention). QUIC's built-in mutual authentication via TLS 1.3 satisfies the handshake precondition. CAPABILITY_EXCHANGE begins on the first application-level datagram after QUIC connection establishment.

C.3 WebSockets

Discovery manifests are transmitted as WebSocket binary frames containing CBOR-encoded messages. No additional length prefix is required as WebSocket frames are self-delimiting. Mutual authentication requires an application-level handshake prior to capability exchange, as WebSockets do not natively support mutual TLS in all deployment environments. The Noise Protocol Framework IK handshake pattern is RECOMMENDED for this purpose.

C.4 Local Wireless Broadcast (Open Radio)

Discovery broadcasts are transmitted as unencrypted UDP datagrams to the local broadcast or multicast address. Each datagram contains exactly one CBOR-encoded message with no length prefix. Broadcasts MAY include the full discovery manifest or MAY use lightweight discovery (peer_id and personal_hash only). Session establishment (HANDSHAKE and CAPABILITY_EXCHANGE) occurs over a unicast connection initiated after discovery. Authentication MUST be established before CAPABILITY_EXCHANGE on the unicast channel.

Appendix D. Conformance Levels

Two conformance levels are defined.

D.1 Minimal Conforming Implementation

- A minimal conforming implementation MUST implement:
- CBOR wire format with deterministic encoding per §5.1 and §5.2
- All fields defined in §4.1 (top-level manifest fields), including protocol_version
- Personal hash computation over deterministic CBOR per §4.5
- The four-message capability exchange sequence defined in §7.2, including CONFIRM reconciliation
- Role-based intersection logic per §7.4
- NONE outcome handling — session MUST close on NONE (§7.6)
- Rebroadcast on all normative triggers (§8.1)
- The built-in UNRESTRICTED and DENIED classification tiers (§6.1)
- Sequence number replay detection per §4.1
- Glare resolution per §7.1.3
- core.session.heartbeat.1 per Appendix B.1

D.2 Full Conforming Implementation

- A full conforming implementation MUST implement everything in D.1, plus:
- Peer cache with auto-deny per §9
- Classification resolution and trust gate evaluation per §6
- Scope parameter enforcement (rate_limit, ttl, and any registered extensions)
- Parameter negotiation via CONFIRM accepted_params per §7.3
- Active set continuity during re-exchange per §8.4
- COSE credential verification per §5.4
- Build attestation exchange per core.session.attest.1 (Appendix B.2)

Appendix E. Connection Archetypes — Worked Examples

The following examples use the four-message exchange sequence defined in §7.2. All messages are CBOR-encoded per §5; examples are shown in CBOR diagnostic notation for readability.

E.1 NONE / NONE Archetype

Neither peer offers capabilities the other can consume.

```
A -> B : OFFER { 1:1, 2:{1:1, 2:h'A1...', 3:42,
         4:[{1:"core.data.stream.1", 2:1}],
         5:h'FF01...', 6:"sha-256"} }
B -> A : OFFER { 1:1, 2:{1:1, 2:h'B2...', 3:17,
         4:[{1:"core.data.stream.1", 2:1}],
         5:h'FF02...', 6:"sha-256"} }

A computes intersection: PROVIDE + PROVIDE = no match
A -> B : CONFIRM { 1:2, 2:h'FF01...', 3:[] }

B computes intersection: PROVIDE + PROVIDE = no match
B -> A : CONFIRM { 1:2, 2:h'FF02...', 3:[] }

Both peers reconcile: [] ∩ [] = []
Both peers: NONE
A -> B : CLOSE { 1:3, 2:h'FF01...', 3:1 }
Both peers store: { peer_id, last_hash, last_outcome: NONE }
```

E.2 ASYMMETRIC Archetype

A provides a stream capability; B consumes it.

```
A -> B : OFFER { 1:1, 2:{..., 4:[{1:"core.data.stream.1", 2:1,
         5:{2:3600}}], ...} }
B -> A : OFFER { 1:1, 2:{..., 4:[{1:"core.data.stream.1", 2:2}], ...} }

A computes intersection: PROVIDE + CONSUME = match
A evaluates trust gate for peer_B -> ALLOW
A -> B : CONFIRM { 1:2, 2:h'...', 3:["core.data.stream.1"] }

B computes intersection: CONSUME + PROVIDE = match
B evaluates trust gate for peer_A -> ALLOW
B -> A : CONFIRM { 1:2, 2:h'...', 3:["core.data.stream.1"] }

Reconciliation: ["core.data.stream.1"] ∩ ["core.data.stream.1"]
Both peers: ACTIVE (1 capability)
```

E.3 ASYMMETRIC with Trust Disagreement

Both peers compute a match, but their trust gates produce different results.

```
A -> B : OFFER { ..., 4:[
```

```
         {1:"core.data.stream.1", 2:1},
         {1:"core.data.blob.1", 2:1}], ... }
B -> A : OFFER { ..., 4:[
         {1:"core.data.stream.1", 2:2},
         {1:"core.data.blob.1", 2:2}], ... }

A trust gates: both ALLOW
A -> B : CONFIRM { 1:2, 2:h'...', 3:["core.data.blob.1",
         "core.data.stream.1"] }

B trust gates: stream ALLOW, blob DENY
B -> A : CONFIRM { 1:2, 2:h'...', 3:["core.data.stream.1"] }

Reconciliation: intersection = ["core.data.stream.1"]
Both peers: ACTIVE (1 capability)
```

E.4 MUTUAL Archetype with Parameter Negotiation

Both peers provide and consume capabilities simultaneously. Scope parameters are negotiated.

```
A -> B : OFFER { ..., 4:[
         {1:"core.data.stream.1", 2:1, 5:{2:7200}},
         {1:"core.data.blob.1", 2:2, 5:{1:100, 2:3600}},
         {1:"core.session.heartbeat.1", 2:3, 3:true}], ... }

B -> A : OFFER { ..., 4:[
         {1:"core.data.stream.1", 2:2, 5:{2:3600}},
         {1:"core.data.blob.1", 2:1, 5:{1:50, 2:7200}},
         {1:"core.session.heartbeat.1", 2:3, 3:true}], ... }

A computes intersection: 3 matches
A -> B : CONFIRM { 1:2, 2:h'...',
         3:["core.data.blob.1", "core.session.heartbeat.1",
            "core.data.stream.1"],
         4:{"core.data.blob.1": {1:50, 2:3600},
            "core.data.stream.1": {2:3600}} }

B mirrors computation
B -> A : CONFIRM { 1:2, 2:h'...',
         3:["core.data.blob.1", "core.session.heartbeat.1",
            "core.data.stream.1"],
         4:{"core.data.blob.1": {1:50, 2:3600},
            "core.data.stream.1": {2:3600}} }

Reconciliation: 3-way match.
  core.data.blob.1:              rate_limit=min(100,50)=50, ttl=min(3600,7200)=36
```

```
  core.data.stream.1:            ttl=min(7200,3600)=3600
```

E.5 Rebroadcast and Re-evaluation

A is ACTIVE with B. A removes core.data.stream.1 from its capability list.

```
A recomputes personal_hash (hash_A2, sequence_num++)
A -> B : OFFER { 1:1, 2:{..., 3:43, 4:[
         {1:"core.data.blob.1", 2:2},
         {1:"core.session.heartbeat.1", 2:3, 3:true}],
         5:h'hash_A2', ...} }

B observes hash_A2 != hash_A -> re-enters CAPABILITY_EXCHANGE
Note: existing active set (3 capabilities) remains operational

B -> A : OFFER { ..., 4:[...], ... }

A -> B : CONFIRM { 1:2, 2:h'hash_A2',
         3:["core.data.blob.1", "core.session.heartbeat.1"] }
B -> A : CONFIRM { 1:2, 2:h'...',
         3:["core.data.blob.1", "core.session.heartbeat.1"] }

Reconciliation: 2-way match.
core.data.stream.1: deactivated
core.data.blob.1: not interrupted
core.session.heartbeat.1: not interrupted
Session remains ACTIVE with 2 capabilities.
```

E.6 Auto-Deny

A has previously completed a NONE exchange with B. B reappears with same hash.

```
B broadcast received by A (CBOR: {2:h'B2...', 5:h'FF02...'})
A checks peer cache: peer_id_B found
  last_hash == current hash_B : true
  last_outcome == NONE        : true

A records outcome: DENIED
A does NOT initiate handshake

--- Later: B updates its capabilities, rebroadcasts with hash_B2 ---

A checks peer cache: peer_id_B found
  last_hash (hash_B) != current hash (hash_B2) : true
```

```
A transitions peer_B from DENIED to PEER_VISIBLE
A initiates new handshake -> CAPABILITY_EXCHANGE
```

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This document is a work in progress. Author contact information to be added prior to publication.